

Libraries, spaCy & Tokens

Post-Assessment 1 · 11 questions · 38 points

Name: _____

Date: _____

Score: / 38

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

In every spaCy question, assume this code has already run: `import spacy` and `nlp = spacy.load("en_core_web_sm")`

1. What is a library, really?

4 points

Meanings:

1. A single `.py` file of code you can import.
2. A folder of modules grouped together and installed as one unit.
3. A big collection of code for one kind of job, like spaCy.
4. The set of functions and classes a library lets you use, and the rules for calling them.

(a) Which meaning number matches **package**? (1 pt)

(b) Which meaning number matches **library**? (1 pt)

(c) In `from datetime import date`, what is `datetime`? (1 pt)

(d) In `from datetime import date`, what is `date`? (1 pt)

2. Installing libraries safely

3 points

(a) Write the pip command that installs the `spacy` library.

(1 pt)

(b) Write the pip command that lists every package installed in the environment.

(1 pt)

(c) Why should each project get its own virtual environment?

(1 pt)

3. Why language is hard

2 points

Read the sentence: **"I saw the man with the telescope."**

Describe **two** different meanings this sentence could have.

4. Hello, spaCy

3 points

```
doc = nlp("Robots can dance!")
print(len(doc))
print(doc[1].text)
```

(a) What is each item inside a `Doc` called?

(1 pt)

(b) What does `print(len(doc))` show?

(1 pt)

(c) What does `print(doc[1].text)` show?

(1 pt)

5. Tokenization: beyond split()

4 points

```
text = "They didn't go, see kidcode.org!"
print(len(text.split()))
print([t.text for t in nlp(text)])
```

(a) What does the first `print` show?

(1 pt)

(b) Write spaCy's tokens in order, separated by `|`.

(2 pts)

(c) Name one thing spaCy's tokenizer handles better than `.split()` in this sentence.

(1 pt)

6. Blank or trained?

2 points

(a) You need to find the nouns and verbs in a story. Which line is best?

(1 pt)

(A) `nlp = spacy.blank("en")`

(B) `nlp = spacy.load("en_core_web_sm")`

(b) What is inside a pipeline made with `spacy.blank("en")`, and what is missing?

(1 pt)

7. Token Sorter

5 points

Write **T** (True) or **F** (False) in every box. Each row is one token.

Token	is_alpha	is_digit	like_num	is_punct	like_url
Leo					
7					
three					
!					
www.kidcode.org					

8. Doc indexing

3 points

`doc = nlp("A brave knight saved the tiny village.")`. Write what each expression gives.

Expression	Result
<code>doc[2].text</code>	
<code>doc[4:7].text</code>	
<code>doc[-1].text</code>	

9. Clean-Up Crew

4 points

Fill in the blanks so `clean` keeps the **lowercase** text of every token that **is alphabetic** and is **not** a stop word. Then write what it prints.

```
doc = nlp("A Happy dog runs to THE park.")
clean = [t._____ for t in doc if t._____ and not t._____]
print(clean)
```

(a) Write the three missing attribute names, in order.

(3 pts)

(b) What does it print?

(1 pt)

10. Word Family Finder

3 points

spaCy reads: "The geese flew south while my cat is sleeping."

(a) Write the `lemma_` of each word.

(2 pts)

Word	geese	flew	is	sleeping
<code>lemma_</code>				

(b) Why do you need a trained model (not `spacy.blank`) to get good lemmas?

(1 pt)

11. Cleaning recipe

5 points

Write a function `clean_words(text)` that runs `nlp` on the text and uses a **list comprehension** to **return** the lowercase text of every token that is alphabetic and not a stop word.

EXAMPLE: `CLEAN_WORDS("DOGS ARE BARKING LOUDLY, AREN'T THEY?")` RETURNS

```
['dogs', 'barking', 'loudly']
```

END OF ASSESSMENT